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Roll no. 53-FET/BSCE/F23

Subject:- System Programming.

Assignment No. 2

```
# include <stdlib.h>
```

```
# include <stdio.h>
```

```
# include <unistd.h>
```

```
struct msg-buffer {
```

```
    long msg-type;
```

```
    int number;
```

```
};
```

```
long long calculate_factorial(int n){
```

```
    if (n < 0) return 0;
```

```
    long long fact = 1;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        fact *= i;
```

```
    }
```

```
    return fact;
```

```
}
```

```
int main() {
```

```
    int msg_id;
```

```
    struct msg-buffer message;
```

```
    printf("Enter the message queue ID provided by client:");
```